



THE KING'S SCHOOL

Depth Study – Year 9 Science

PART A - SCIENTIFIC REPORT

TITLE: *PREDICTION OF THE AMOUNT OF POTASSIUM IODIDE ON REACTION SPEED IN THE DECOMPOSITION OF HYDROGEN PEROXIDE WITH REGRESSION NEURAL NETWORKS*

INTRODUCTION / SCIENTIFIC BACKGROUND OF TOPIC:

Hydrogen peroxide ($2\text{H}_2\text{O}_2$) is a reactive substance that possesses a peroxide bond which consists of a single oxygen-oxygen covalent bond (O-O). This bond is relatively weak and decomposes easily, especially under exposure to ultraviolet (UV) light, breaking down into water (H_2O) and oxygen gas (O_2) as represented by the following equation:



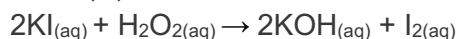
This reaction is exothermic, meaning it releases heat energy as the bonds break, and is often used as an easy and safe way in obtaining oxygen gas. Thus, hydrogen peroxide is used as a safer and more environmentally friendly solution for removing organic waste, such as wastewater treatment, by rapidly decomposing the organic matter in an oxidization reaction. (e.g. Ksibi, 2006)

Role of Potassium Iodide as a Catalyst

To effectively conduct applications such as wastewater treatment, the oxidization process must be done in a reasonable timeframe, meaning there is a need for a sufficient level of oxygen concentration that needs to be maintained in the reaction.

However, the natural decomposition of hydrogen peroxide under UV exposure is both energy and time-intensive, limiting large-scale applications.

To tackle this challenge, potassium iodide (KI) is often used as a catalyst in this decomposition reaction by providing an alternate reaction pathway that requires lower activation energy. Its catalytic mechanisms is described as involving the formation of an intermediate iodine species (I_2), which serves as a key component in accelerating the decomposition reaction of hydrogen peroxide by Margerum et al. (1994). According to them, this process begins when iodide ions (I^-) react with hydrogen peroxide, forming iodine (I_2) and water in an initial oxidation step:



And subsequently, this resulting iodine reacts with the additional hydrogen peroxide, resulting in the regeneration of iodide ions and formation of water and oxygen gas, and adhering to the law of conservation of mass, which states that the total mass of elements in the beginning of a chemical reaction will equal the total mass of elements in the end of the chemical reaction.

This catalytic cycle allows potassium iodide to continuously facilitate the reaction without being consumed, making it a practical and efficient catalyst to accelerate the decomposition of hydrogen

peroxide in large scale. Another study by Whitehead et al. (2000) have also highlighted how KI is being used as a highly efficient catalyst from wastewater treatment to environmental remediation, where this rapid generation of oxygen is crucially needed.

Chemical Reaction Rate Prediction

Identifying and predicting the change in the rate of reaction is essential for optimizing catalyst-reactant ratio, especially in applications such as wastewater treatment, where high concentrations and failure to remove substances such as potassium iodide could lead to serious health concerns such as food poisoning. Traditionally, this relies on constructing kinematic equations based on experimental data. However, these approaches can either become too complex or inaccurate when dealing with non-linear behaviors and multiple influencing factors. (Fogler, 2016)

Neural Networks

A neural network, as the name suggests, is a computing algorithm that models the anatomy of the human brain's network of neurons. It consists of multiple different layers of neurons which each processes the inputs from the previous layer and outputs to the next. This approach enables computers to execute actions without being explicitly programmed by a human to do so. Depending on the specific neural network architecture, training of a neural network model usually involves pushing large sets of data through these layers of neurons to fine tune variables called "biases" that alters the output. To represent multi-element, high dimensional data sets that are often used to train this network, neural network architects have conventionally taken the approach of converting such data into mathematical tensors, which is similar to a scalar, vector and matrix but can be to the n th dimension. An example of this approach can be seen in the encoder section of Google Brain team's "Transformer" Architecture, which established the basis for modern large language models such as ChatGPT "Generative Pretrained **Transformer**". (Vaswani et al., 2017)

Regression Neural Network Architecture

The regression neural network, in particular, is a neural network architecture that enables the prediction of future data points based on inspecting the pattern of past data results. The training of such model is done by fitting the curve of a function into points of a data set, and predicting future elements through evaluating through such function in locations unproven by data. The regression neural network architecture is suitable for the context of this study due to its exceptional ability to predict continuous variables and low requirement for large sets of training data. In this context of chemical kinematics, past research papers have demonstrated how a regression neural network trained on experimental data outperformed traditional models in precision. (Yin et al., 2021)

Application in this study

In this study, a regression neural network model is developed based on experimental data on the decomposition reaction of hydrogen peroxide (H_2O_2) catalyzed by varying concentrations of potassium iodide catalyst (KI). By training implementing this model, the study seeks to identify optimal catalyst (KI) to reactant (H_2O_2) ratio that maximizes reaction efficiency, offering a significant insight for effective and efficient applications in an industrial perspective.

AIM:

To investigate the effect of varying amount of potassium iodide (KI) on the decomposition rate of hydrogen peroxide (H_2O_2).

VARIABLES:

Independent Variable: Amount of potassium iodide

Dependent Variable: Volume of oxygen created

Controlled Variables (and how you will keep them the same):

- 1) Concentration of Hydrogen Peroxide (H_2O_2):
 - Use different measuring cylinder for hydrogen peroxide to prevent unwanted reaction and influencing foam production.
- 2) Concentration of Potassium iodide (KI):
 - Use different measuring cylinder for potassium iodide to prevent unwanted reaction and influencing foam production.
- 3) Flask size and shape:
 - Use the same size and shape of the measuring cylinder in each trial to prevent variations in surface area available for the reaction which can influence the rate of oxygen release and foam production.
- 4) Cleanliness of measuring cylinder for reaction:
 - Thoroughly clean and dry the measuring cylinder for reaction with water in each trial to prevent remaining water and chemicals from influencing the measurements and reaction rate.
- 5) Type and amount of detergent:
 - Use 1 ml of the same detergent and mix thoroughly to prevent too much or differing detergent from influencing the foam production.

HYPOTHESIS:

It is predicted that within the 1-minute timeframe, if the amount of potassium iodide catalyst is increased, the volume of generated oxygen gas will increase in a linear pattern.

RISK ASSESSMENT:

Equipment/ Substance/Process	Possible hazard and risk to person	Precautions to manage these hazards and risks to a person(s)
Hydrogen Peroxide (H_2O_2)	Ingestion or inhalation causing irritation on operator's skin and respiratory system.	- wear protective glasses to prevent spilling into eyes - place closed on a flat surface on the lab bench. - if in eyes, immediately contact science staff and rinse with clean water.
Potassium Iodide (KI)	Contact and ingestion causing significant eye and skin irritation	- wear protective glasses and clothing to prevent chemical contacting exposed skin. - place closed on a flat surface on the lab bench. - if in eyes, immediately contact science staff and rinse with clean water.

Measuring Cylinder	Chipped and shattered glass causing cuts to operator.	- place on a flat surface and away from the edge of the lab bench. - if shattered, immediately notify science staff and do not attempt to pick up shards with bare hand. - wear hard, enclosed footwear at experiment site
Chemical Foaming	Foam overflow and spilled chemicals can cause slippery floors, contaminate the workspace and potentially expose the operator to chemicals	- Use appropriately sized flasks for facilitating the reaction and containing chemicals. - Place all containers of chemicals on a flat surface and away from the edge. - if spilled, immediately notify science staff and do not touch the containments with bare hand.

EQUIPMENT/ MATERIALS:

ITEM NAME	CLEAR DESCRIPTION AND NUMBER REQUIRED
Measuring Cylinder (2x10ml)	2 x 10ml measuring cylinders made of plastic or glass to accurately measure chemicals
Measuring Cylinder (100ml)	1 x 100ml measuring cylinder made of glass to facilitate the reaction.
Hydrogen Peroxide (3% wt/wt)	1 x 150ml of Hydrogen peroxide (H_2O_2) 3% wt/wt concentration that will be used consistently across all experiment trials.
Potassium Iodide 1.0mol/L	1 x 60ml of Potassium Iodide (KI) 1.0mol/l concentration that will be used across all trials as the catalyst
Detergent	1 x 15ml of the same detergent used to form foam that allows the measuring of oxygen gas produced in the reaction
Pipette (2ml)	1 x 2ml pipette used to measure detergent
Stopwatch	1 x Stopwatch used to measure the timeframe for the reaction.

METHOD:

- Preparation of Hydrogen Peroxide (2% wt/wt):
 - 10mL of Hydrogen Peroxide (2% wt/wt) was added into one 10mL measuring cylinder
- Preparation of Potassium Iodide (1.0mol/L):
 - 2mL of potassium iodide (1.0mol/L) was added into another 10mL measuring cylinder which is different from the one used for Hydrogen Peroxide
- Combining Hydrogen Peroxide:
 - The 10 mL of Hydrogen Peroxide was transferred into a 100 mL measuring cylinder.
- Addition of Detergent:
 - 1 mL of detergent was added to the same 100 mL measuring cylinder and mixed thoroughly by gently swirling the cylinder in a circular motion to ensure even distribution.
- Recording Initial Volume:**
 - The total volume of the current mixture (Hydrogen Peroxide and detergent) was recorded. This volume was then added to the volume of Potassium Iodide used in the reaction, and the sum was recorded as

the initial total volume of chemicals before the reaction.

6. Initiating the Reaction:

- The 2 mL of Potassium Iodide (1.0 mol/L) was quickly added to the 100 mL measuring cylinder containing the Hydrogen Peroxide and detergent mixture. Simultaneously, the stopwatch was started.

7. Measuring Foam Production:

- After 1 minute, the highest point where the foam displayed a solid color was observed and recorded. The volume increase of the mixture was calculated by subtracting the initial total volume of chemicals (before the reaction) from the final foam volume and recorded.

8. Cylinder Cleaning:

- The contents of the 100 mL measuring cylinder were disposed of in a suitable chemical waste container. The cylinder was then thoroughly washed under running tap water to remove any residual chemicals.

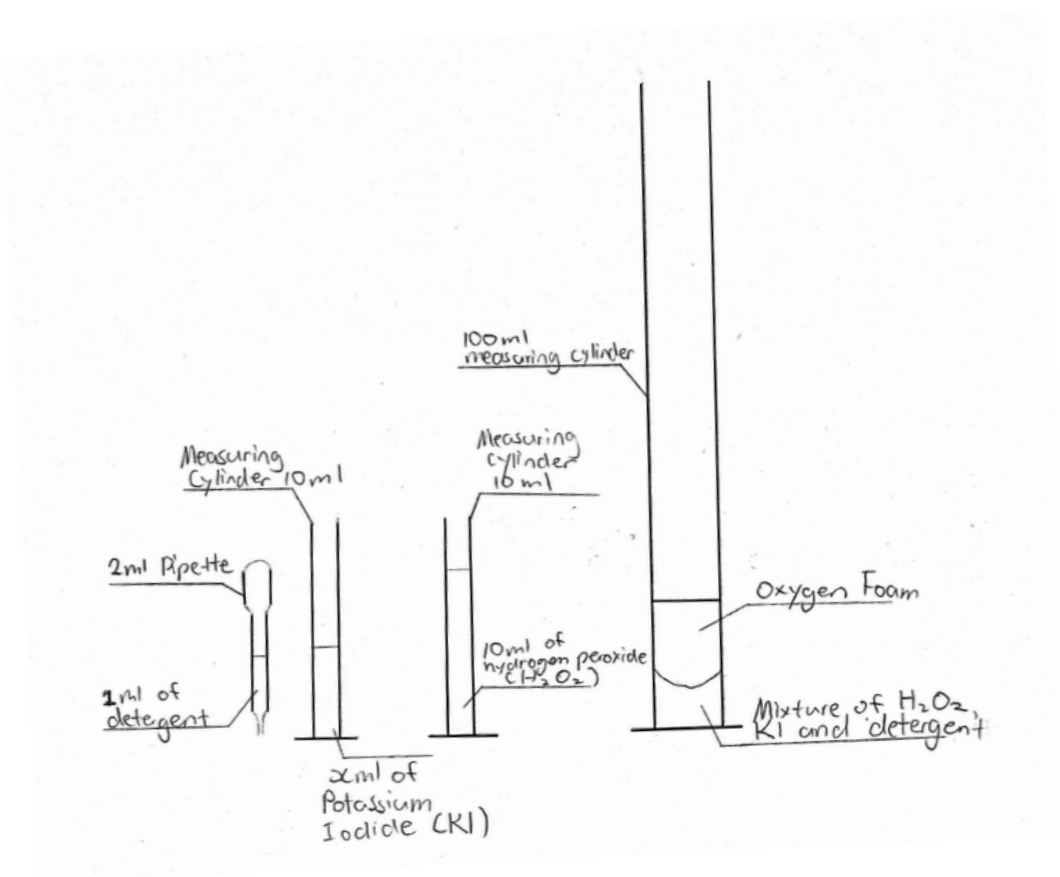
9. Drying the Cylinder:

- Remaining droplets of water were removed by gently swinging the 100 mL measuring cylinder, causing the water droplets to be expelled from the cylinder.

10. Repetition of Trials:

- Steps 1 through 9 were repeated using 3 mL, 4 mL, 5 mL, and 6 mL of Potassium Iodide (1.0 mol/L) for three trials each. Additionally, two more trials were conducted with 2 mL of Potassium Iodide. For each trial, the volume increase of the mixture was recorded.

PHOTOGRAPH / LABELLED SCIENTIFIC DIAGRAM:



RESULTS:

Qualitative Observations/Diagram of your results (if appropriate)

1. Initial Reaction Behavior

Upon adding the potassium iodide catalyst to the mixture of hydrogen peroxide and detergent, an immediate and rapid increase in foam production was observed, indicating the quick release of oxygen gas. This reaction also appeared to be quite vigorous, with visible bubbling

at the edge between the mixture and foam.

2. Effect of Increasing Potassium Iodide (KI)

As the amount of added potassium iodide increased, it was observed that the intensity and rate of foam production also increased. Simultaneously, the reaction also became progressively more vigorous, with foam reaching high levels in the cylinder more quickly.

3. Foam Characteristics

The produced form was dense and maintained a uniform structure throughout the experiment, without collapsing significantly. The foam appeared to be yellow visually, and gave off a faint stench, possible due to the produced oxygen gas.

4. Temperature variation

After the reaction, touching the outer surface of the measuring cylinder where the reaction took place with an exposed hand can feel a noticeable increase in the temperature of the mixture near the base of the measuring cylinder compared to before the reaction took place. Additionally, As the amount of added potassium iodide increased, this temperature change seemed to also be amplified, suggesting that higher catalyst concentrations could potentially result in a more significant exothermic effect.

Table of Results:

Correlation Between Potassium Iodide Concentration and Foam Production in the Catalytic Decomposition of Hydrogen Peroxide				
	Produced foam volume increase (mL)			
Amount of Potassium Iodide (mL)	Trial 1	Trial 2	Trial 3	Average (2 Significant Figures)
2	14	12	11	12
3	14	13	15	14
4	28	22	19	23
5	27	35	28	30
6	31	32	34	32

Graph of Results:

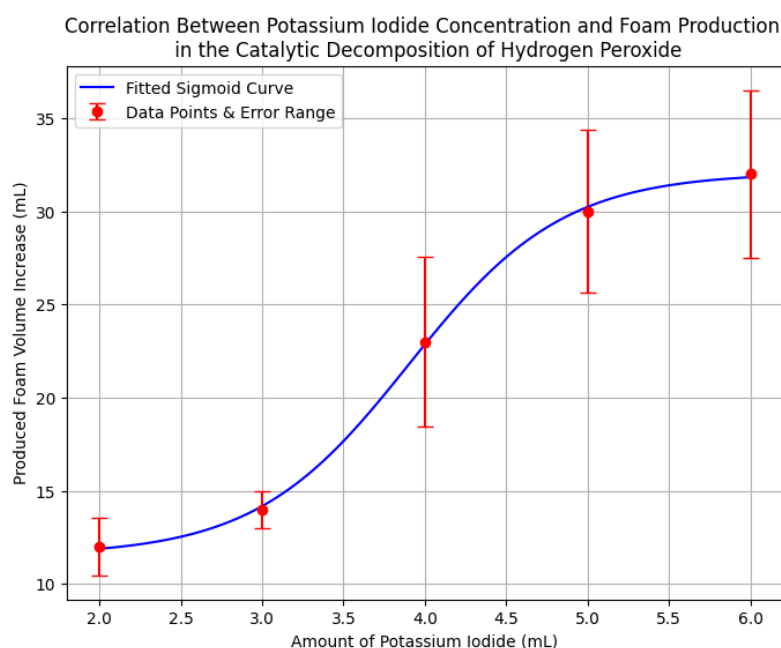


Figure 1 - Correlation Between Potassium Iodide Concentration and Foam Production in the Catalytic Decomposition of Hydrogen Peroxide

Analysis of Regression Neural Network Model

As seen in figure 1, it is observable that the experimental data exhibit a sigmoidal pattern. This is characterized by the small rate of foam increase at both low and high amount of potassium iodide catalyst, but a stronger rate of foam increases in the intermediate amount of added potassium iodide. Thus based on this pattern, an open-source machine learning library named Scikit-learn will be used to fit a sigmoidal curve to the experimental data set.

$$y = \frac{L}{1 + e^{-k(x-x_0)}} + b$$

Where:

y – produced foam volume increase
x – amount of potassium iodide (KI)
L – Curve's maximum value
 x_0 – value of x at midpoint of sigmoid
k – steepness of the curve
b – baseline value when x = 0

Training Outcome (2 d.p.):

L = 20.54
 x_0 = 3.90
k = 2.12
b = 11.53

This pattern gives several implications regarding the nature of this trend:

1. Initial Mixture Volume (Baseline - b):

The constant b = 11.53mL represents the rough total volume of the mixture in the 100mL measuring cylinder before the reaction took place. This adheres to the experimental mixture volume of hydrogen peroxide and detergent of 11mL.

2. Maximum Foam Production (Maximum Value - L):

The constant L = 20.54 represents the approximate upper bound for the volume of produced foam. In the experiment, this is likely the point where adding more potassium iodide catalyst will be unable to significantly increase the foam produced without adding more hydrogen peroxide reactant.

3. Optimal Catalyst Amount (Midpoint of Sigmoid - x_0):

The equation x = 3.90 is the predicted location for the centre of the sigmoidal trend, where the addition of Potassium Iodide (KI) catalyst yields the most efficient results. Beyond this point, the effect of adding more potassium iodide will start to diminish, as the foam production approaches the maximum value L.

This model has demonstrated that the optimal amount of potassium iodide in the KI catalysed decomposition of hydrogen peroxide is 3.9 mL of KI catalyst per 10.53 mL (11.53 – 1 mL detergent) of hydrogen peroxide, or a ratio of **1 : 27**.

The source code used to train the model can be found here:

https://github.com/LycoKodo/KodoBase/blob/992173afc8d81fe71db11e139e58b05f27c7a1bf/24%20Science%20SRP%20Source%20Code/model_train/train.py

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